Model Predictive Controller for Position Control of Twin Rotor MIMO System

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***Abstract*— Twin Rotor Multi-Input Multi-Output System (TRMS) is a highly non-linear 2-DOF helicopter model. It consists of two rotors, vertical and horizontal, whose positions need to be controlled with the help of suitable controller. Due to its non-linear characteristics, it is quite challenging to design a traditional controller for this system. The design objectives of the controller are to stabilize the positions of both the rotors of TRMS in presence of cross coupling between the two rotors and reach the desired Pitch and Yaw angles. Model predictive control (MPC) can be suitably used in such multivariable systems with significant constraint handling. In this paper, model predictive control (MPC) is used for the regulation of Pitch (vertical angle) and Yaw (horizontal angle). The controller is implemented on the linearized model of TRMS. The code is generated based on the model, controller and optimisation equations. The weight matrices, control horizon, output horizon, reference and the matrices generated from MATLAB coding are used as input to the MATLAB Simulation as well as to the real-time Simulation. The results obtained from m-file, Simulation and the Real-time for various references are also included in this paper.**

Keywords*— *Twin Rotor MIMO System, Non-Linear, Multivariable, 2-DOF, Pitch, Yaw, Position control, Linear Model Predictive Controller, constraints, quadprog, m-file, Simulation, Real-time implementation..

I. INTRODUCTION

A non-linear system, Twin Rotor MIMO System (Feedback Instruments), is considered for the implementation of Model Predictive Control. TRMS, being a coupled system and having certain amount of constraints on its inputs, is quite difficult to control. But MPC has many advantages over traditional controllers, which make it suitable for being used to control the position of the rotors of such a complicated system.

A MATLAB code, satisfying the constraints on the inputs and the input increments (in presence of noise and disturbances), is implemented. The simulation is developed with the help of the weight matrices, control horizon, output horizon, reference and the matrices generated from the m-file. The command written to represent the system model in the m-file is replaced by the non-linear system model in the simulation. Whereas in real-time implementation, the system model is replaced by the actual hardware set up blocks. The tracking results are studied for various references under certain disturbances and noise included. The detailed explanation is done in the respective sub-topics.

II. DESCRIPTION AND MODELLING OF TRMS

A. Lab Set-up Description:

Twin Rotor MIMO System [1] is a laboratorial set-up of a helicopter model with a few remarkable simplifications. Unlike in real helicopter, the position of TRMS is controlled through the rotor velocity variations. Whereas in real helicopter, the propulsion is varied through the rotor blades angle modification. But, the dynamic characteristics such as cross coupling (the change in any one rotor/axis position will affect the other) between the two axes and highly non-linear behavior of a helicopter are present in the TRMS. The difference between the TRMS and helicopter subjected to various points is well explained in Table 1.

The TRMS is controlled with two inputs, u_1 and u_2 (the voltages to the front and rear rotors respectively). There is a PCI connected to the computer that looks after the inter-connections between the PC and the TRMS. Whereas measured signals are transferred via I/O card (ADC/DAC). The mechanical and the electrical unit together make it a control system setup. There is a Control unit which is used to turn on/off the supply to the electrical counterpart as shown in Fig. 1.

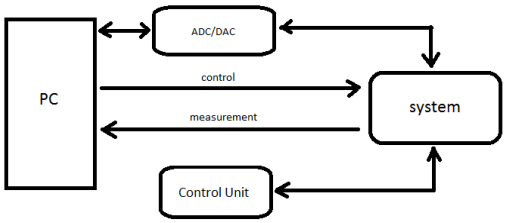


Fig. 1. TRMS Schematic Hardware Setup

TABLE 1 MAIN DIFFERENCE BETWEEN TRMS AND HELICOPTER

Constraints	TRMS	Helicopter
Ability to fly	No	Yes
Pivot point is located at	Midway between the two rotors	The main rotor head
Pitch control	Speed control of the main rotor	Collective pitch control
Yaw control	Speed control of tail rotor speed	Angle of the blades of the tail rotor
Cyclical control	No	Yes

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B. Model:

The mechanistic model of the TRMS is presented in Figure 2 along with the momentum as well as the pitch and yaw position movement representation. According to the electro-mechanical diagram given in Fig. 2., the non-linear model equations can be derived. The momentum equations derived for the vertical and the horizontal movements are as follows [1]:

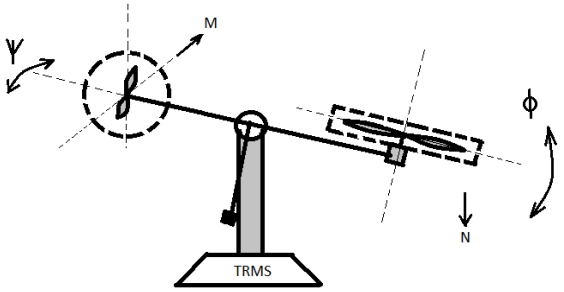


Fig. 2. TRMS mechanistic model

1) Vertical movement:

Using conservation of momentum, the equation obtained for the pitch movement is given by equation 1,

$$I_1.\ddot{\alpha}_v = M_1 - M_{FG} - M_{B\alpha_v} - M_G \tag{1}$$

Where,

M_1 - Non-linear static characteristics for front propeller

$$M_1 = a_1.\tau_1^2 + b_1.\alpha_1 \tag{2}$$

M_{FG} - Momentum due to gravity

$$M_{FG} = M_g.\sin\alpha_v \tag{3}$$

$M_{B\alpha_v}$ - Frictional Forces momentum

$$M_{B\alpha_v} = B_{1\alpha v}.\dot{\alpha}_v - B_{2\alpha_v}.\sin 2\alpha_v.\dot{\alpha}_h^2 \tag{4}$$

M_G - Gyroscopic momentum

$$M_G = k_{gy}.M_1.\dot{\alpha}_h.\cos\alpha_v \tag{5}$$

2) Horizontal movement:

Similarly using the conservation of momentum for yaw movement, we obtain equation 6,

$$I_2.\ddot{\alpha}_h = M_2 - M_{B\alpha_h} - M_R \tag{6}$$

Where,

M_2 - Non-linear static characteristic for rear propeller

$$M_2 = a_2.\tau_2^2 + b_2.\alpha_2 \tag{7}$$

$M_{B\alpha_h}$ - Frictional forces momentum

$$M_{B\alpha_h} = B_{1\alpha h}.\dot{\alpha}_h \tag{8}$$

M_R - Cross-reaction momentum

$$M_R = K_c \frac{T_0 s + 1}{T_p s + 1}.M_1 \tag{9}$$

The two DC motors with the electrical control circuits are approximated by a first order transfer function for each motor. In Laplace domain, the motor momentums for pitch and yaw are given by equation 10 and equation 11 respectively.

$$\tau_1 = \frac{K_1}{T_{11}s + T_{10}}.u_v \tag{10}$$

$$\tau_2 = \frac{K_2}{T_{21}s + T_{20}}.u_v \tag{11}$$

The approximate values of model parameters (chosen experimentally) are given in Table 2.

TABLE 2 TRMS PHYSICAL PARAMETERS

Symbol	Description	Value
I_1	Moment of inertia of vertical rotor	$6.8*10^{-2}$ Kg-m^2
I_2	Moment of inertia of horizontal rotor	$2*10^{-2}$ Kg-m^2
a_1	Static characteristic parameter	0.0135
b_1	Static characteristic parameter	0.0924
a_2	Static characteristic parameter	0.02
b_2	Static characteristic parameter	0.09
M_g	Gravity momentum	0.32N-m
$B_{1\alpha v}$	Friction momentum parameter	$6*10^{-2}$N-m-s^2/rad
$B_{2\alpha v}$	Friction momentum parameter	$1*10^{-5}$N-m-s^2/rad
$B_{1\alpha h}$	Friction momentum parameter	$1*10^{-1}$N-m-s^2/rad
$B_{2\alpha h}$	Friction momentum parameter	$1*10^{-2}$N-m-s^2/rad
k_{gy}	Gyroscopic momentum parameter	0.05s/rad
k_1	Main motor gain	1.1
k_2	Tail motor gain	0.8
T_{11}, T_{10}	Main motor denominator parameters	1.1, 1 resp.
T_{21}, T_{20}	Tail motor denominator parameters	1, 1 resp.
T_p, T_0	Cross reaction momentum parameters	2, 3.5 resp.
k_c	Cross reaction momentum gain	-0.2

C. Linearized Model:

The State matrices can be obtained by using the model of the TRMS in Simulink. The Simulink model is run separately without any input and output. Then using the "linmod" command in the command window we obtain the state matrices. There is a requirement of rearranging the rows and columns in accordance with our input and output considerations [2].

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$$A=\begin{bmatrix} -0.833 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1.246 & -4.706 & -0.089 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1.482 & 0 & 0 & 3.6 & 0 & -5 & 18.75 \\ -0.017 & 0 & 0 & 0 & 0 & 0 & -0.5 \end{bmatrix}$$

$$B=\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad C=\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}, \quad D=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

We define the state variables as,

$x_1 = \tau_1$ -the momentum of main motor,
$x_2 = \alpha_v$ -the pitch angle,
$x_3 = \Omega_v$ -the pitch angular velocity in the vertical plane,
$x_4 = \tau_2$ -the momentum of tail motor,
$x_3 = \alpha_h$ -the yaw angle,
$x_6 = \Omega_h$ -the yaw angular velocity in the horizontal plane
$x_7 = M_R$ -the cross-reaction momentum.

III. CONTROLLER

A. Model Predictive Control

Model Predictive Control (MPC) is an optimal control methodology which works on numerical optimization concept. Future control inputs and future plant responses are predicted using a system model. The optimization of the control input is obtained iteratively with respect to a performance index. This performance index is subjected to some specified constraints. The origin of MPC technology as a computational technique for improving control performance within the process and petrochemical industries has now extended to a wide variety of application areas including chemicals, vehicle path planning and control, food processing, underwater vehicle guidance, missile guidance, Spacecraft rendezvous with space station (fuel optimality), automotive, aerospace, and other such large-scale operation control problems with multiple control variables. MPC has a sound theoretical basis and its stability, optimality, and robustness properties are well understood. Moreover, a linear MPC can be used to control non-linear systems.

MPC delivers by keeping the constraints on inputs, change in input and states in account. Such constraints are present in all the control engineering applications. The constraints show the restrictions on switches, valves, actuators and plant states which are due to the economic, physical, or safety limitations. This makes it a safer controller for hardware implementation.

B. Equations

A common discrete state space model is given by [6]

$$x(k+1) = Ax(k) + Bu(k) \tag{12}$$

$$y(k) = Cx(k) + Du(k) + d(k) \tag{13}$$

where, x(k+1) represents the plant's state; u(k) represents it inputs; y(k) represents its outputs and d(k) represents disturbances with D=0. Assume n_x states, n_u control horizon, n_y prediction horizon. Discrete models are one-step ahead prediction models, that is, given data at sample 'k', one can determine data at sample 'k+1'.

$$y(k+1) = Cx(k+1) + d(k+1) \tag{14}$$

Using the value of $x(k+1)$ from equation 12, we can write equation 14 as,

$$y(k+1) = CAx(k) + CBu(k) + d(k+1) \tag{15}$$

Normally one would assume that the best guess gives:

$$d(k+1) = d(k) \tag{16}$$

Hence, a generalized expression of 1-step ahead prediction is given by equation 17 and equation 18.

$$x(k+1) = Ax(k) + Bu(k) \tag{17}$$

$$y(k+1) = CAx(k) + CBu(k) + d(k) \tag{18}$$

And a generalized expression for the n-step ahead prediction is given by equation 19 and equation 20.

$$x(k+n) = A^n x(k) + A^{n-1}Bu(k) + A^{n-2}Bu(k+1) + \cdots + ABu(k+n-2) + Bu(k+n-1) \tag{19}$$

The system output can be determined simply using,

$$y(k+n) = CA^n x(k) + CA^{n-1}Bu(k) + A^{n-2}Bu(k+1) + \cdots \cdots + ABu(k+n-2) + Bu(k+n-1) + d(k) \tag{20}$$

It is convenient to separate predictions into the part which is known and the part which is yet to be determined. Thus, separating the known and unknown parts gives:

$$\underline{x}(k+1) = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^n \end{bmatrix} x(k) + \begin{bmatrix} A & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{n-1}B & A^{n-2}B & \cdots & B \end{bmatrix} \begin{bmatrix} u(k\,|\,k) \\ u(k+1\,|\,k) \\ \vdots \\ u(k+n-1\,|\,k) \end{bmatrix}$$

This equation can be written as:

$$\underline{x}(k+1) = P_x x(k) + H_x \underline{u}(k) \tag{21}$$

where, P_x, H_x are the Model parameters or Prediction parameters; $x(k)$ are the measurements and $\underline{u}(k)$ are the decision variables.

$$\underline{y}(k+1) = \begin{bmatrix} CA \\ CA^2 \\ \vdots \\ CA^n \end{bmatrix} x(k) + \begin{bmatrix} CA & 0 & \cdots & 0 \\ CAB & CB & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ CA^{n-1}B & CA^{n-2}B & \cdots & CB \end{bmatrix} \begin{bmatrix} u(k\,|\,k) \\ u(k+1\,|\,k) \\ \vdots \\ u(k+n-1\,|\,k) \end{bmatrix} + \begin{bmatrix} d(k) \\ d(k) \\ \vdots \\ d(k) \end{bmatrix}$$

The above equation can be written as:

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$$\underline{y}(k+1) = Px(k) + Ld(k) + H\underline{u}(k) \quad (22)$$

where, P_x, H_x, L are the Model parameters or Prediction parameters; $x(k)$ are the measurements; $d(k)$ is disturbance and $\underline{u}(k)$ are the decision variables. Therefore, the matrices required are P, H, L, P_x, H_x. These matrices depend upon the model parameters A, B, C and the horizon n (assume D=0).

C. System constraints and Performance Index:

The real-time system is subjected to input constraints that restrict the over increment of the input voltage in order to avoid damage to the physical system. The constraints considered are $u_{min} = -2.5$ and $u_{max} = +2.5$ for both the rotor inputs as mentioned in the system manual. These can be expressed in the form of an equation with matrix vectors as:

$$c_c u \le d_d \quad (23)$$

The performance index representing the control objective is given as:

$$J = e^T Qe + u^T Ru + d_u^T T d_u \quad (24)$$

where Q, R, T are the positive definite weight matrices for the tracking error, control input and change in control input respectively. Tracking error is given as $e = r - y$ where $y = Px$.

Tuning of the controller can be done by adjusting the values of these weight matrices, the control horizon and the output horizon along the input constraint consideration. Minimising these performance equations subjected to the constraints mentioned above, gives the control input iteratively. This can be done with the help of quadratic programming by reducing the above performance index to a standard equation.

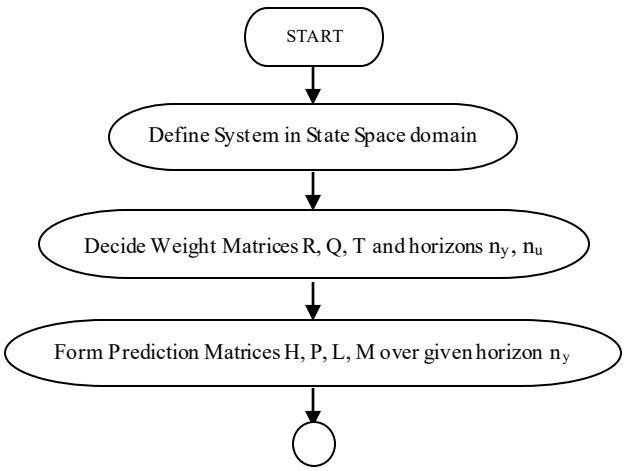
$$J = \tfrac{1}{2} u^T Su + u^T a \quad (25)$$

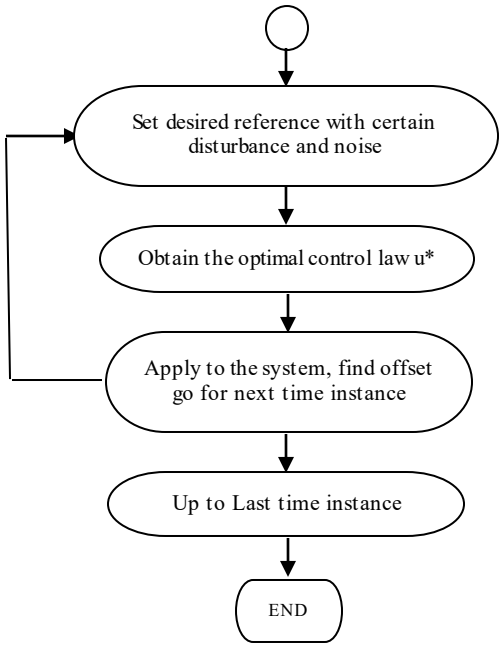
where,

$$S = H^T QH + R + M^T TM\ ,\ a = -2(H^T Qr - H^T QPx) \quad (26)$$

where r is the given reference.

D. Coding Considerations:





IV. RESULTS

A. Coding:

The results obtained after giving various references are shown in the following figures. Each figure consists of four sub-figures. First figure shows reference tracking, second shows the control input required for the set point tracking, third represents the change in control input during the set point tracking and the last one represents the disturbance/noise applied.

1) Step reference with disturbance/noise

A step reference is given to both Pitch and Yaw of 0.4 and 0.5 units respectively with disturbance and noise at some specified instances. The results obtained are given in Figure 3 and Figure 4 for Pitch and Yaw respectively.

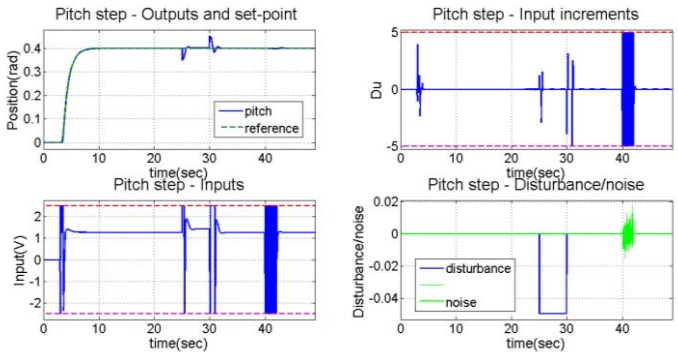


Fig. 3. Pitch: Step reference with disturbance/noise

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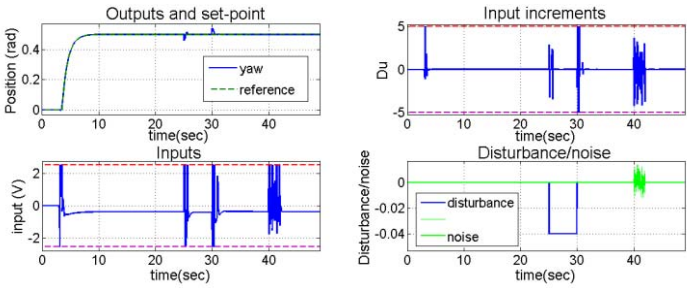


Fig. 4. Yaw: Step reference with disturbance/noise

2) Sine wave reference with disturbance/noise

A randomly generated sine wave reference is given to both Pitch and Yaw respectively with similar disturbance and noise as given in step reference. The results obtained are given in Figure 5 and Figure 6 for Pitch and Yaw respectively.

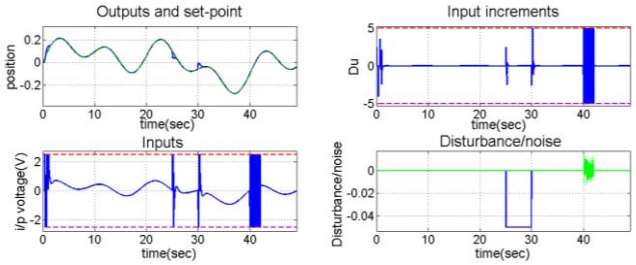


Fig. 5. Pitch: Sine wave reference with disturbance/noise

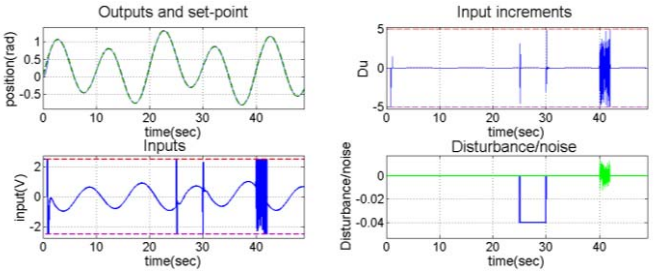


Fig. 6. Yaw: Sine wave reference with disturbance/noise

3) Delayed Sine wave reference with dist/noise

A delayed sine wave reference is given to both Pitch and Yaw respectively with similar disturbance and noise as given in the cases above. The reason to include the delay is to note the difference in tracking in both the cases. In case of delayed input, the error is less as compared to the case without delay. The results for delayed sine reference are given in Figure 7 and Figure 8 for Pitch and Yaw respectively.

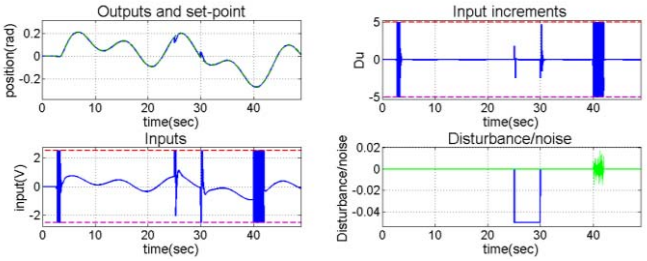


Fig. 7. Pitch: Delayed Sine wave reference with disturbance/noise

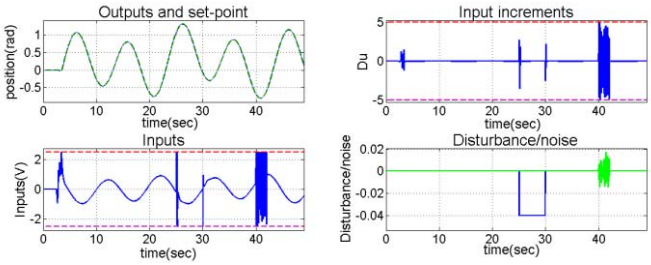


Fig. 8. Yaw: Delayed Sine wave reference with disturbance/noise

B. Simulation:

The Simulation is shown in Figure 9. It consists of a MATLAB function block, TRMS model, Pulse generator along with the Scopes, to workspace and constant blocks.

The References are assigned along with the code within the function block itself. The Pulse generator is used to serve the purpose of matching the operation time of the TRMS model with the actual process. Whereas the constant blocks represent the initial pitch and initial yaw angles as shown in the figure. The Simulation results are shown in the Figure 10, Figure 11, Figure 12 and Figure13. In MATLAB coding, the 'quadprog' command is used for optimization of the performance index. This command has restrictions of code generation in real-time system building, which makes it mandatory to replace the command by writing the optimization equations in the MATLAB code.

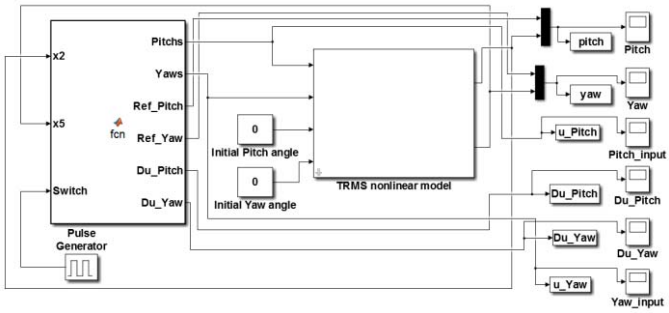


Fig. 9. Simulation

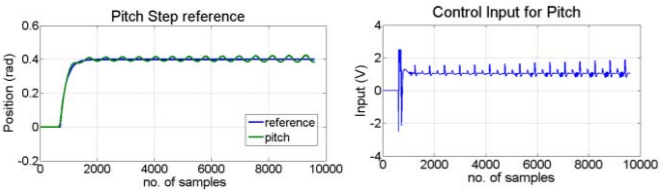


Fig. 10. (a)Pitch Step response, (b)Pitch control input

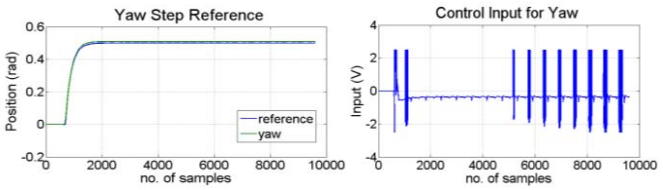


Fig. 11. (a)Yaw Step response, (b)Yaw control input

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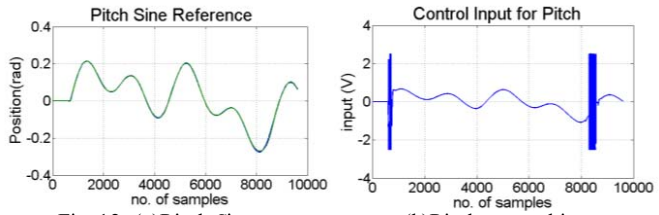


Fig. 12. (a)Pitch Sine wave response, (b)Pitch control input

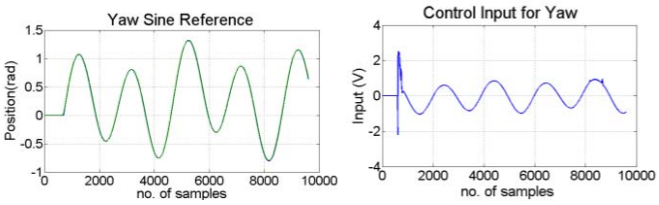


Fig. 13. (a)Yaw Sine wave response, (b)Yaw control input

C. *Real-time Simulation:*

The Real-time Simulation diagram is shown in Figure 14. It consists of the same blocks which are used in Simulation, but the noteworthy difference is that the TRMS model block is replaced with the actual hardware interfaceable TRMS blocks. The Real-time Simulation results are shown in the Figure 15, Figure 16, Figure 17 and Figure 18. The hardware results are better as compared to the Simulation results. Yaw has a better response as compared to Pitch in all the cases considered.

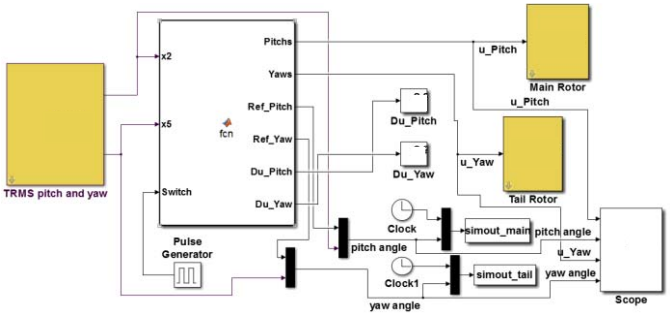


Fig. 14. Real-time Simulation

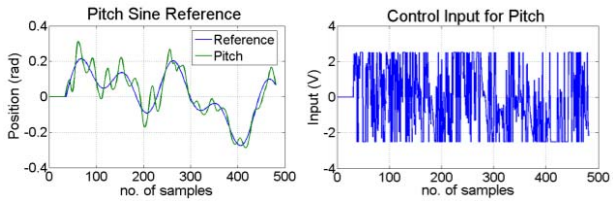


Fig. 15. (a)Pitch Sine wave response, (b)Pitch control input

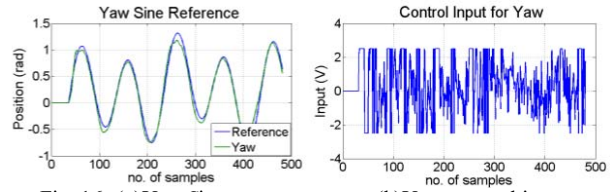


Fig. 16. (a)Yaw Sine wave response, (b)Yaw control input

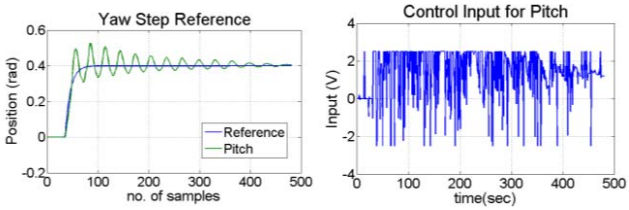


Fig. 17. (a)Pitch Step response, (b)Pitch control input

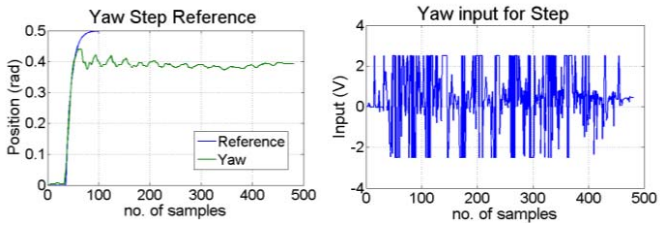


Fig. 18. (a)Yaw Step response, (b)Yaw control input

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